# ECE 3317

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# Review 1

# Complex Vectors & EM theories

- Complex vectors and vector calculus
- Time domain ← → phasor domain
- Maxwell's equations
- Poynting theorem

# Time Domain ← → Phasor Domain

$$V(t) = V_0 \cos(\omega t + \phi)$$

$$\begin{cases} \mathrm{V} = V_0\, e^{j\phi} & \text{going from time domain to phasor domain} \\ V(t) = \mathrm{Re}\left\{ \mathrm{V} e^{j\omega t} \right\} & \text{going from phasor domain to time domain} \end{cases}$$

Time-domain ⇔ Phasor domain $\boxed{V(t) \Leftrightarrow \mathrm{V}}$

For vectors:

$$\underline{V}(t) = \underline{\hat{\mathbf{x}}} V_x \cos(\omega t + \phi_x) + \underline{\hat{\mathbf{y}}} V_y \cos(\omega t + \phi_y) + \underline{\hat{z}} V_z \cos(\omega t + \phi_z)$$

$$\updownarrow \qquad \updownarrow \qquad \updownarrow$$

$$\underline{\mathrm{V}} \equiv \underline{\hat{\mathbf{x}}} V_x e^{j\phi_x} + \underline{\hat{\mathbf{y}}} V_y e^{j\phi_y} + \underline{\hat{\mathbf{z}}} V_z e^{j\phi_z}$$

# Maxwell's Equations

in time domain | | in phasor domain

$$\nabla \times \underline{E} = -\frac{\partial \underline{B}}{\partial t}$$

Faraday's law

$$\nabla \times \underline{\mathrm{E}} = -j\omega \underline{\mathrm{B}}$$

$$\nabla \times \underline{H} = \underline{J} + \frac{\partial \underline{D}}{\partial t}$$

Ampere's law

$$\nabla \times \underline{\mathrm{H}} = \underline{\mathrm{J}} + j\omega \underline{\mathrm{D}}$$

$$\nabla \cdot \underline{B} = 0$$

Magnetic Gauss law

$$\nabla \cdot \underline{\mathrm{B}} = 0$$

$$\nabla \cdot \underline{D} = \rho_v$$

Electric Gauss law

$$\nabla \cdot \underline{\mathrm{D}} = \rho_v$$

# Poynting Theorem

$$-\oiint_S \left(\underline{E} \times \underline{H}\right) \cdot \hat{n}\, dS = \int_V \sigma \left|\underline{E}\right|^2 dV + \frac{\partial}{\partial t} \int_V \left(\frac{1}{2}\mu\left|\underline{H}\right|^2\right) dV + \frac{\partial}{\partial t} \int_V \left(\frac{1}{2}\varepsilon\left|\underline{E}\right|^2\right) dV$$

Power dissipation as heat (Joule's law)

Rate of change of stored magnetic energy

Rate of change of stored electric energy

power flowing into the region $V$.

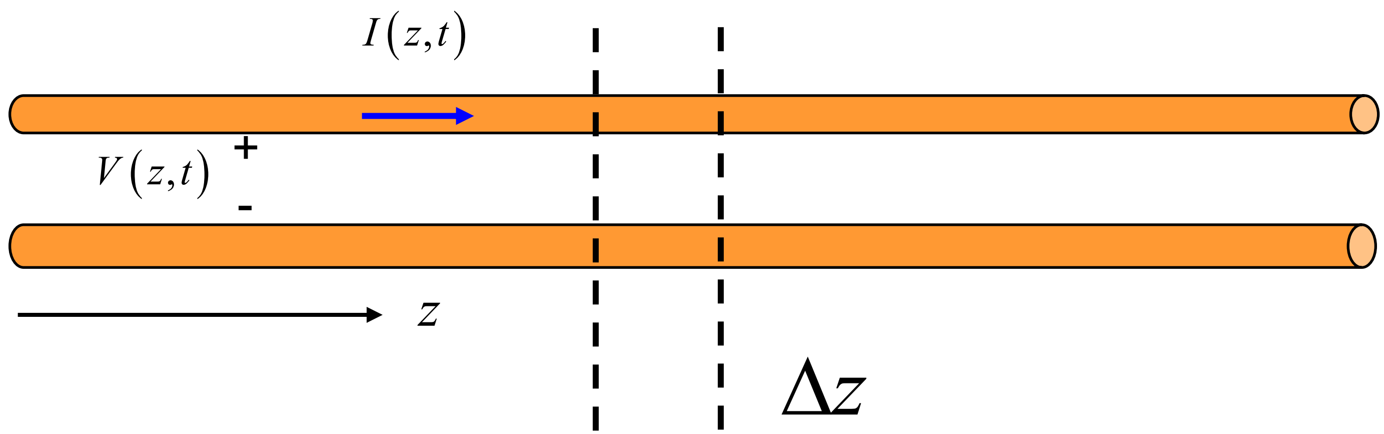
instantaneous Poynting vector: $\boxed{\underline{S}(t) = \underline{E}(t) \times \underline{H}(t)}$

instantaneous power flowing out of the region $= \oiint_S \underline{S}(t) \cdot \hat{n}\, dS$

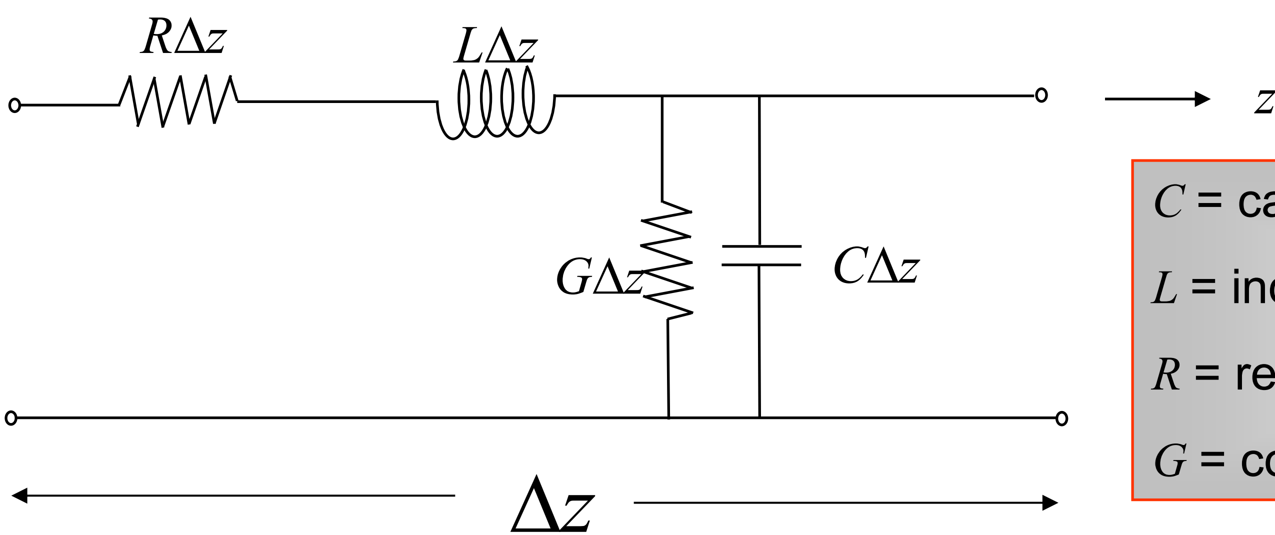
complex Poynting vector: $\boxed{\underline{\mathrm{S}} = \frac{1}{2}\left(\underline{\mathrm{E}} \times \underline{\mathrm{H}}^*\right)}$

time-average power flowing out of the region $= \oiint_S \mathrm{Re}\{\underline{\mathrm{S}}\} \cdot \hat{n}\, dS$

# Circuit Model of Transmission Line




Circuit Model:




$C$ = capacitance/length [F/m]

$L$ = inductance/length [H/m]

$R$ = resistance/length [Ω/m]

$G$ = conductance/length [S/m]

# Equations of Transmission Line

"Telegrapher's Equations (TEs)"

$$\frac{\partial V}{\partial z} = -RI - L\frac{\partial I}{\partial t}$$

$$\frac{\partial I}{\partial z} = -GV - C\frac{\partial V}{\partial t}$$

Lossy wave equation

(no exact solution)

$$\frac{\partial^2 V}{\partial z^2} - (RG)V - (RC + LG)\frac{\partial V}{\partial t} - LC\left(\frac{\partial^2 V}{\partial t^2}\right) = 0$$

Lossless case:

$$R = G = 0$$

"wave equation"

$$\boxed{\frac{\partial^2 V}{\partial z^2} - LC\left(\frac{\partial^2 V}{\partial t^2}\right) = 0}$$

$$\frac{\partial^2 V}{\partial z^2} - \frac{1}{c_d^2}\left(\frac{\partial^2 V}{\partial t^2}\right) = 0 \qquad c_d = \frac{1}{\sqrt{LC}}$$

traveling wave solution

$$V(z,t) = f(z - c_d t) + g(z + c_d t)$$

where $f$ and $g$ are arbitrary functions.

$$c_d = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{LC}}$$

Speed of light in dielectric / transmission line

# Traveling Wave

$$V(z,t) = f(z - c_d t)$$ forward traveling wave

$$V(z,t) = g(z + c_d t)$$ backward traveling wave

characteristic impedance $Z_0$

$$Z_0 = \sqrt{\frac{L}{C}}$$

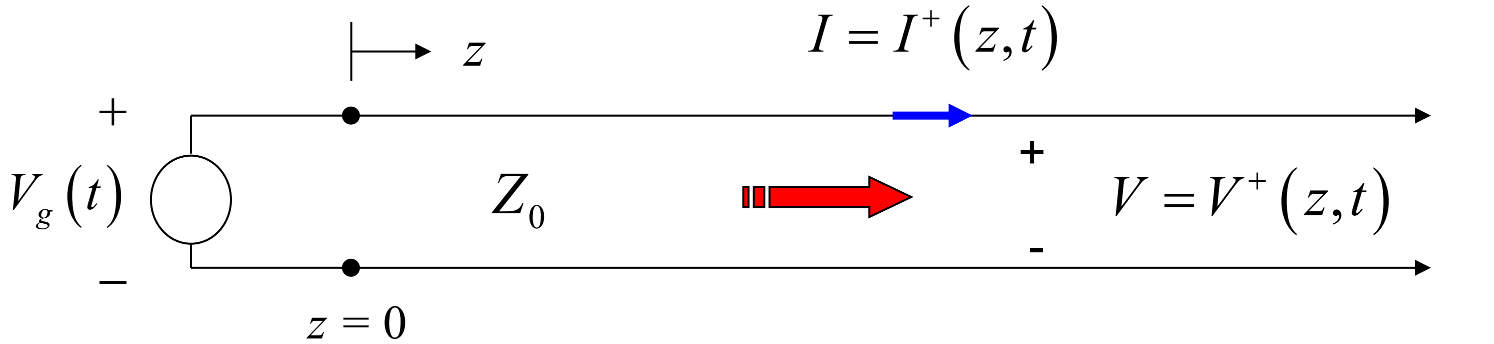
The units of $Z_0$ are Ohms.

$$V(z,t) = f(z - c_d t) + g(z + c_d t)$$

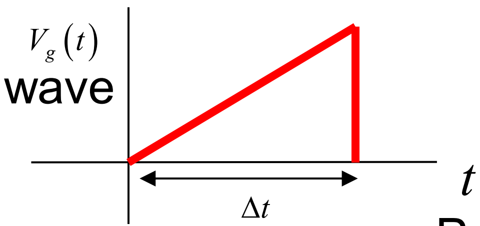
$$I(z,t) = \frac{1}{Z_0}\left[f(z - c_d t) - g(z + c_d t)\right]$$

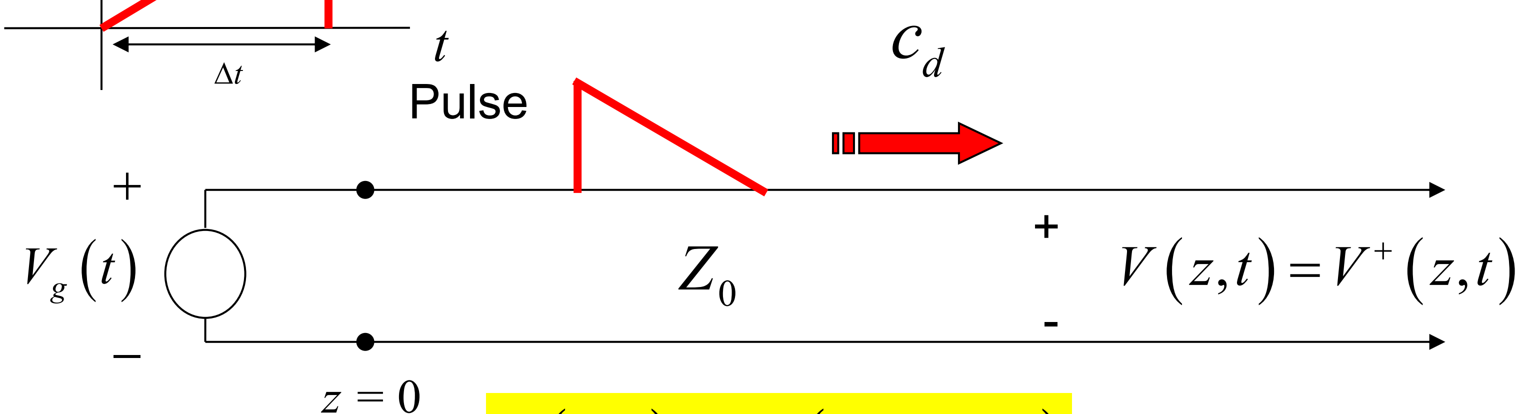
# Wave Propagation in Time Domain

A voltage signal is applied at the input of the semi-infinite transmission line.




Sawtooth wave at source



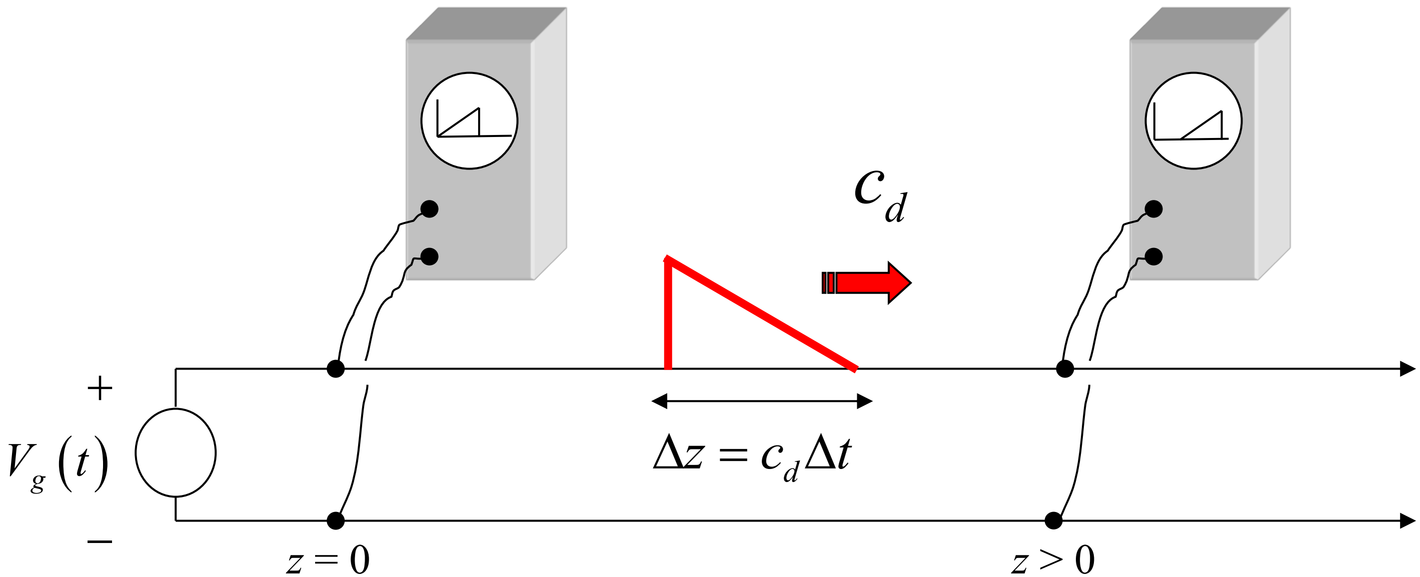



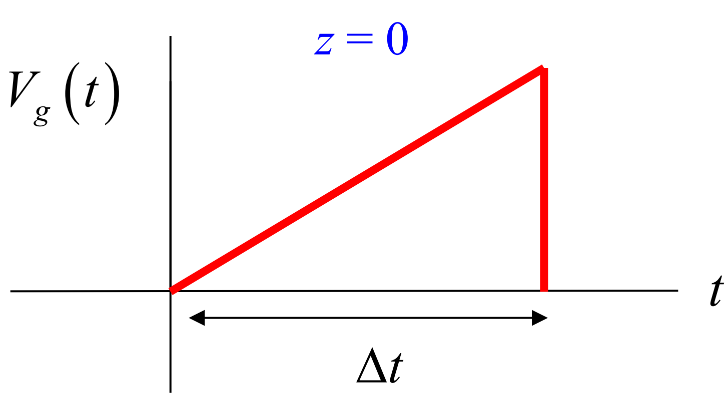

$$V(z,t) = V_g(t - z/c_d)$$

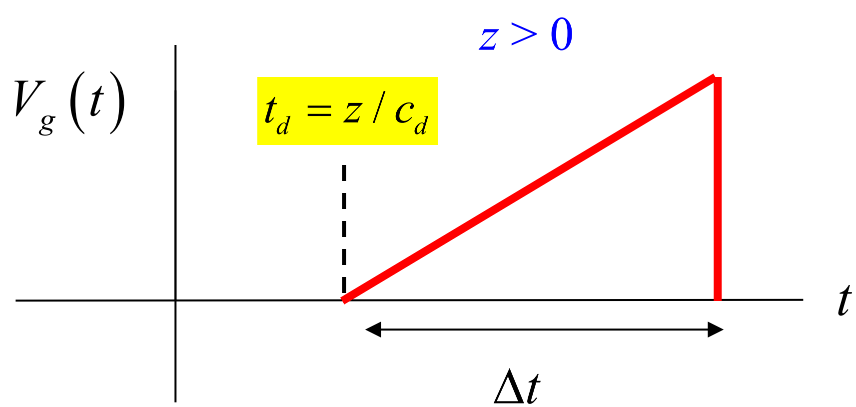
Note that the shape of the pulse as a function of $z$ is a scaled mirror image of the pulse shape as a function of $t$.

# Oscilloscope trace

Note the delay in the trace on this oscilloscope.



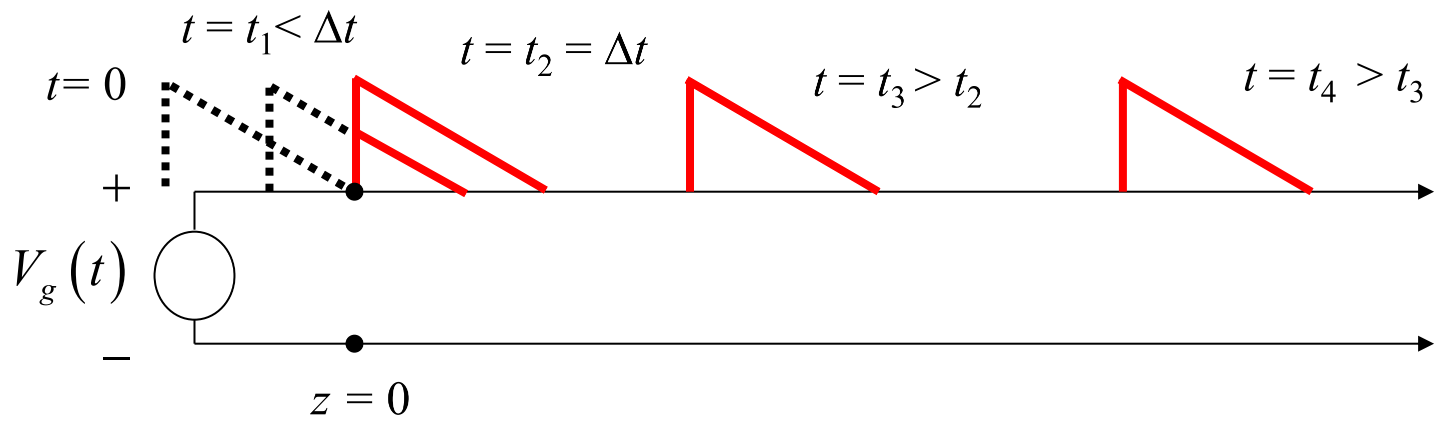



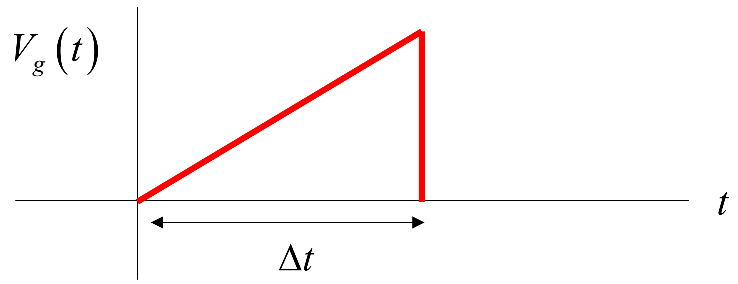



# Snapshots

The pulse is shown emerging from the source end of the line.

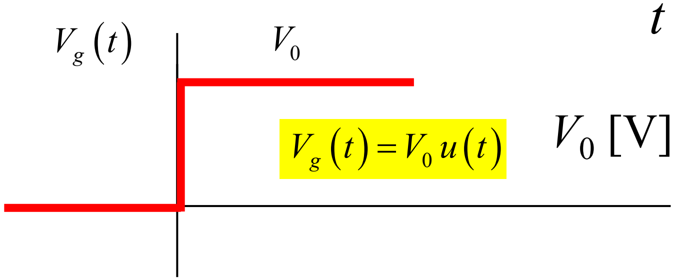
A series of "snapshots" is shown.

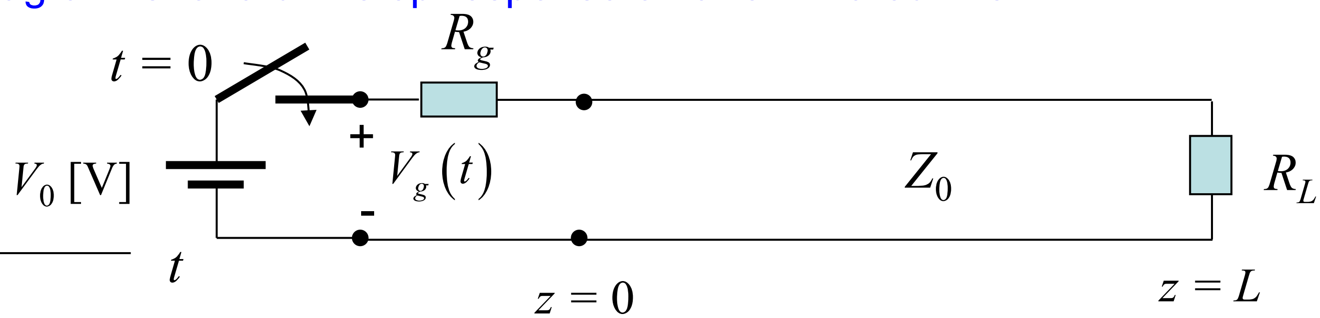


# Bounce Diagram

Bounce diagram is for a unit step response on a terminated line.



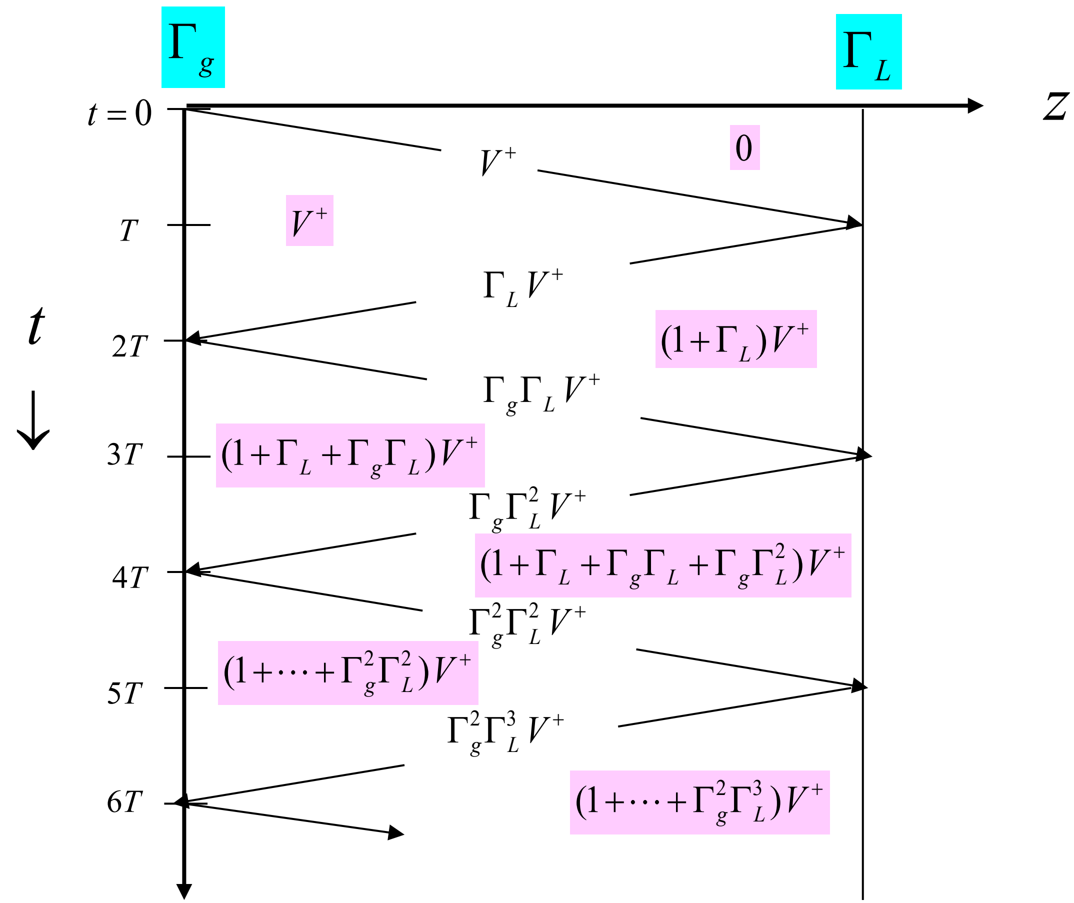




$$T = \frac{L}{c_d}$$

$$V^+ = \left(\frac{Z_0}{R_g + Z_0}\right) V_0$$

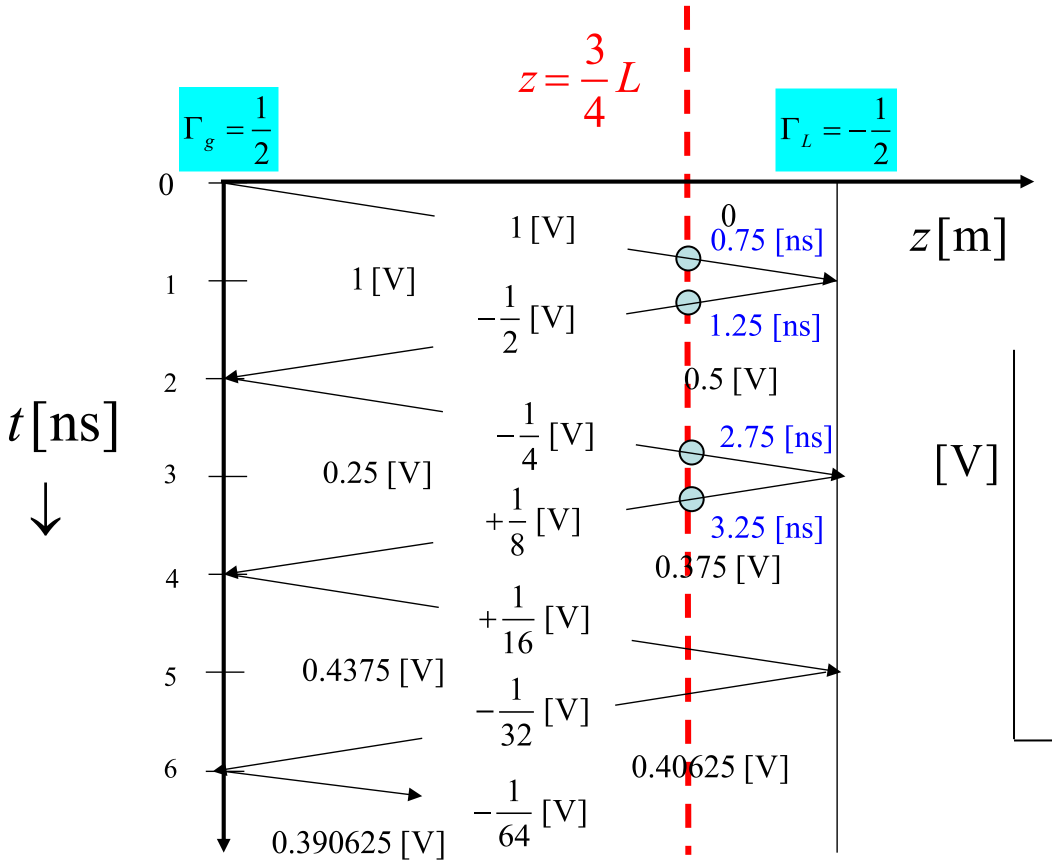
$$\Gamma_L = \left(\frac{R_L - Z_0}{R_L + Z_0}\right)$$

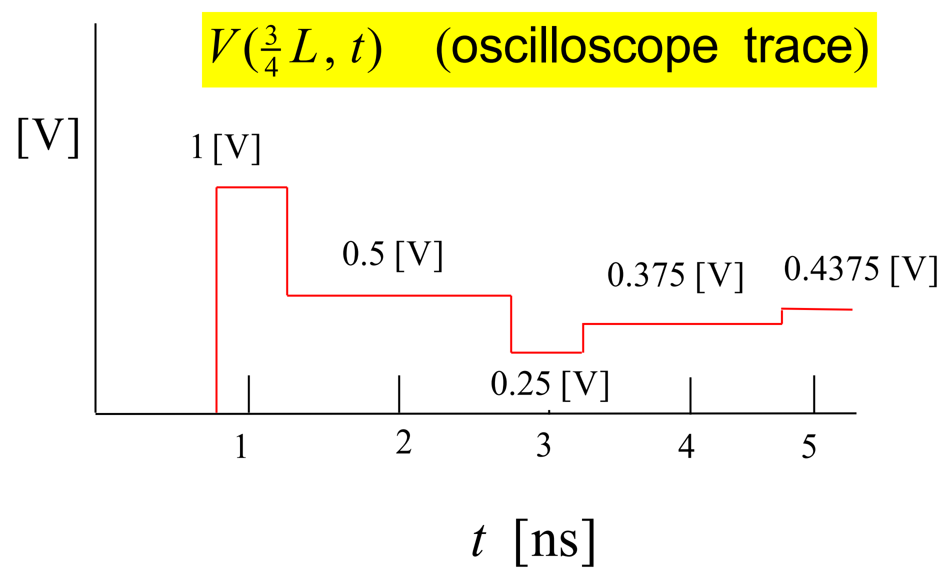
$$\Gamma_g = \left(\frac{R_g - Z_0}{R_g + Z_0}\right)$$



# Oscilloscope Trance

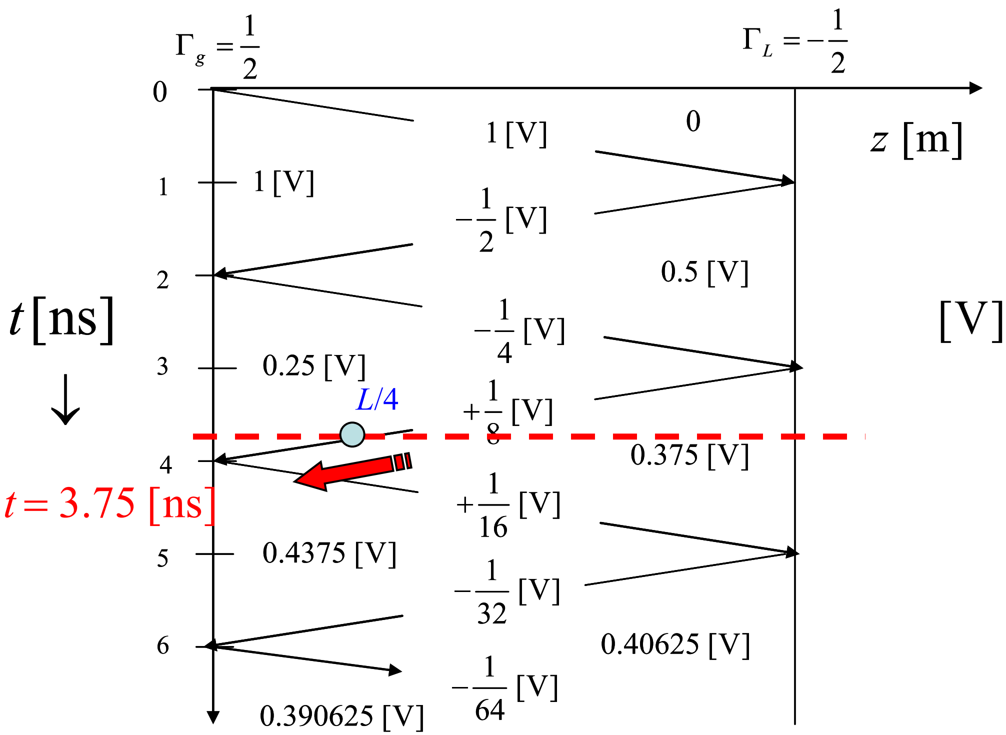
Bounce diagram can be used to get an "oscilloscope trace" at any point on the line.

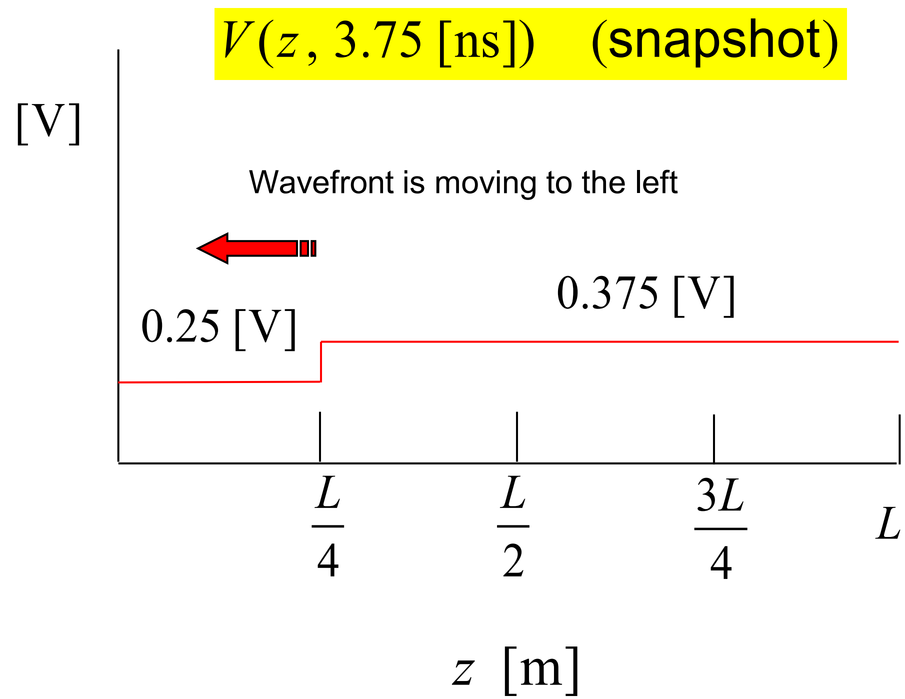


# Snapshot

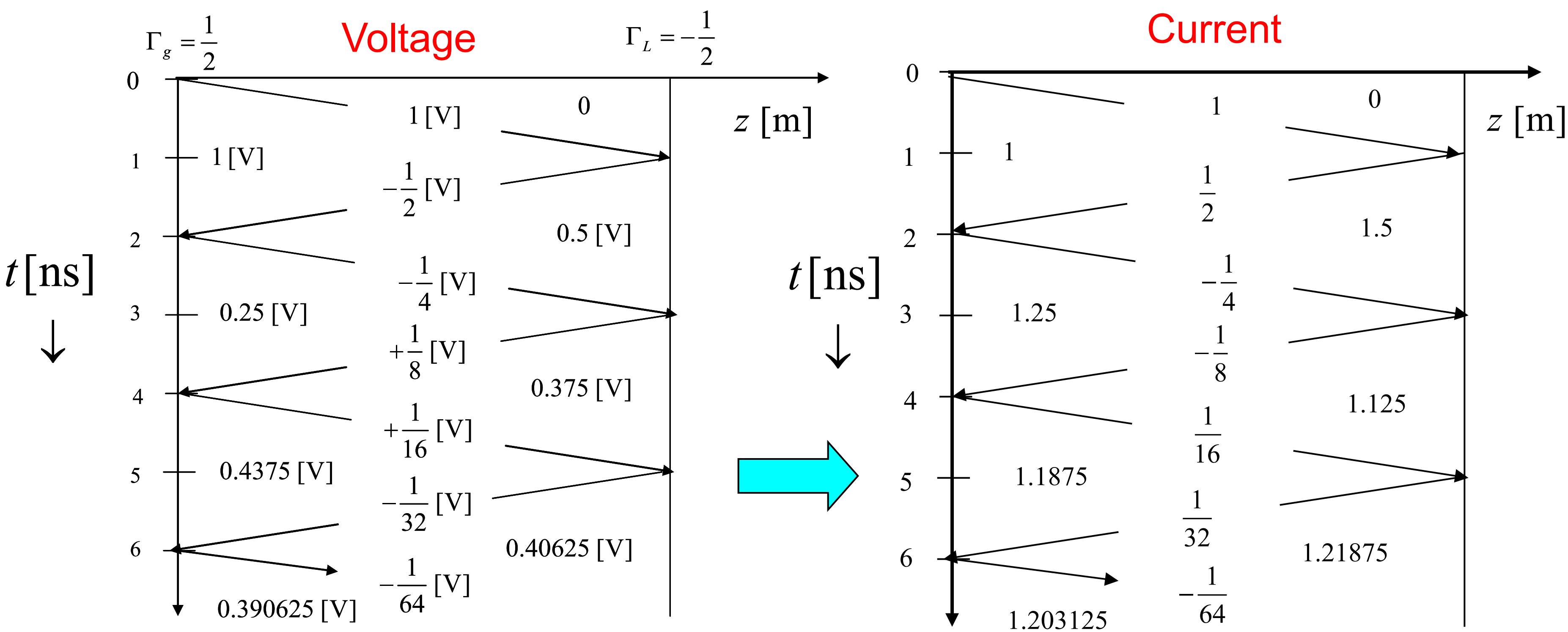
The bounce diagram can also be used to get a "snapshot" of the line voltage at any point in time.



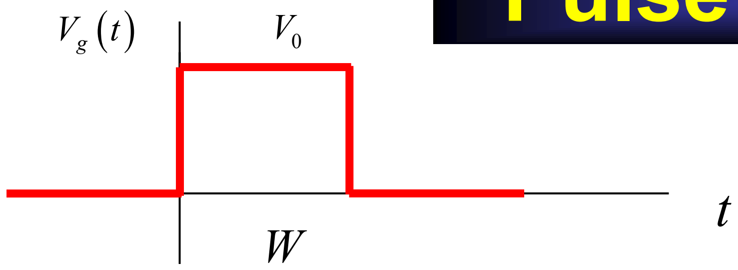
# Bounce Diagram for Current

To obtain a current bounce diagram from the voltage diagram, multiply forward-traveling voltages by $1/Z_0$, backward-traveling voltages by $-1/Z_0$.




Note: This diagram is for the *normalized* current, defined as $Z_0 I(z,t)$.

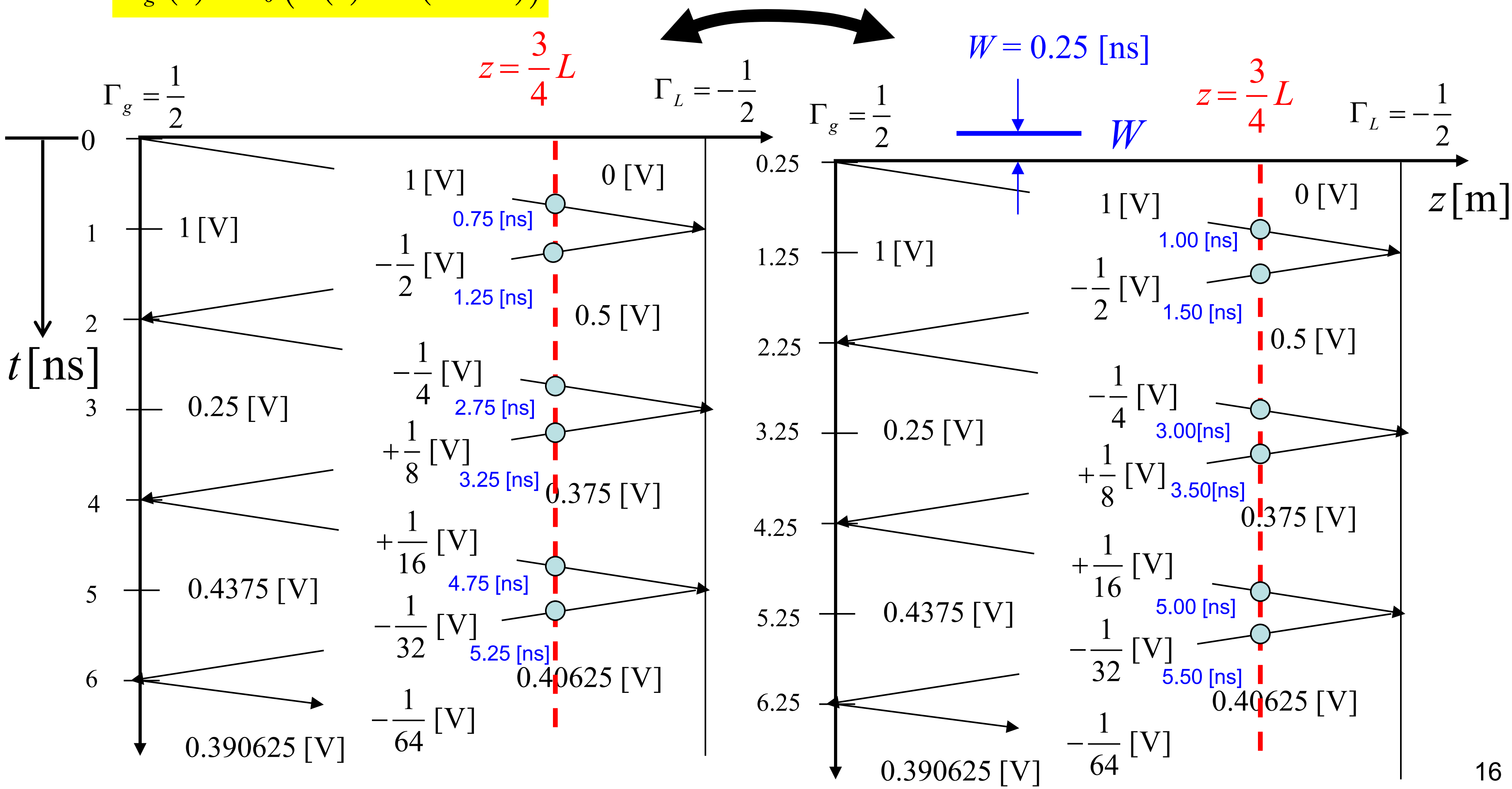
# Pulse Response



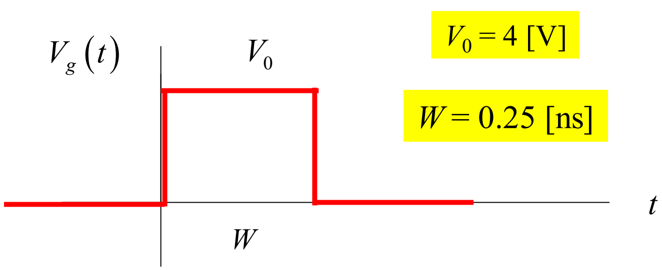

$$V_g(t) = V_0\left(u(t) - u(t-W)\right)$$

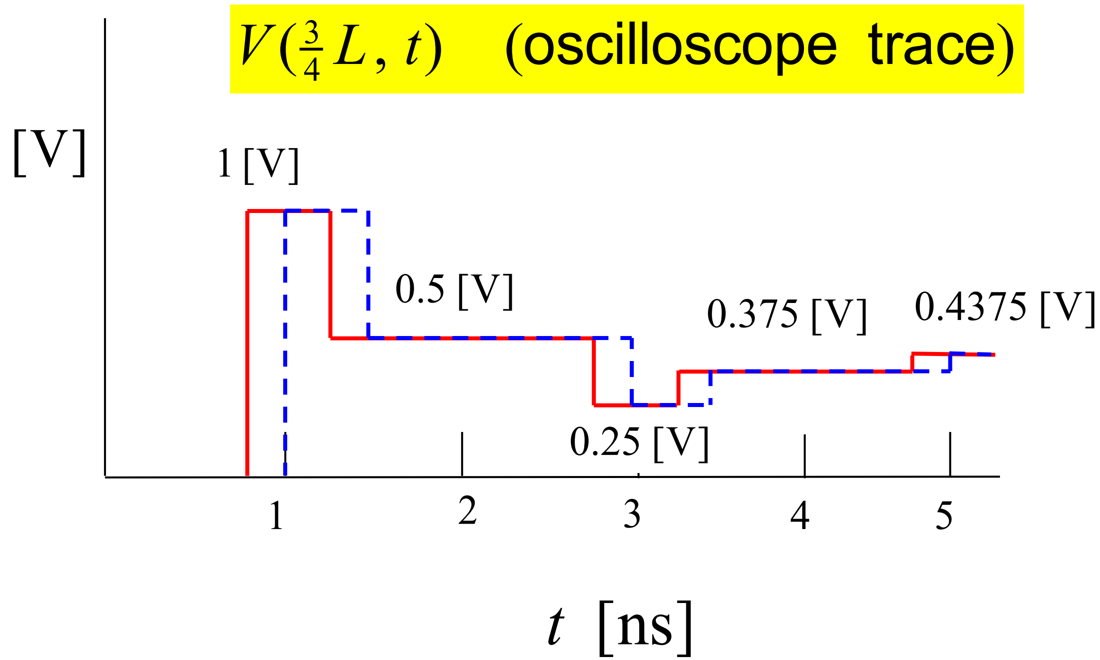
Subtract

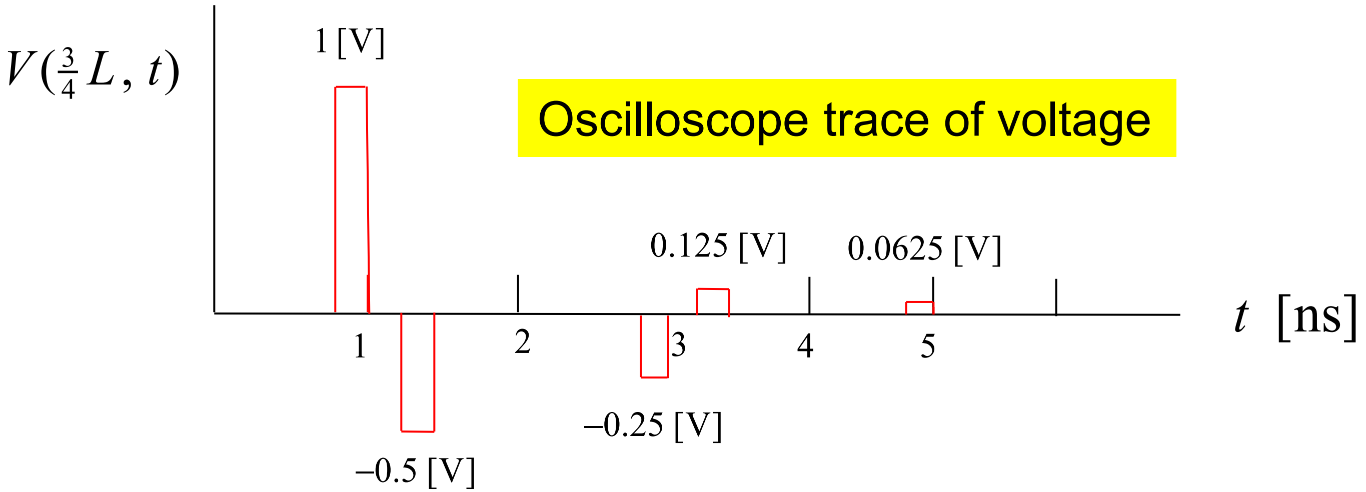
We subtract two bounce diagrams, with the second one being a shifted version of the first one.



# Pulse: Oscilloscope Trace



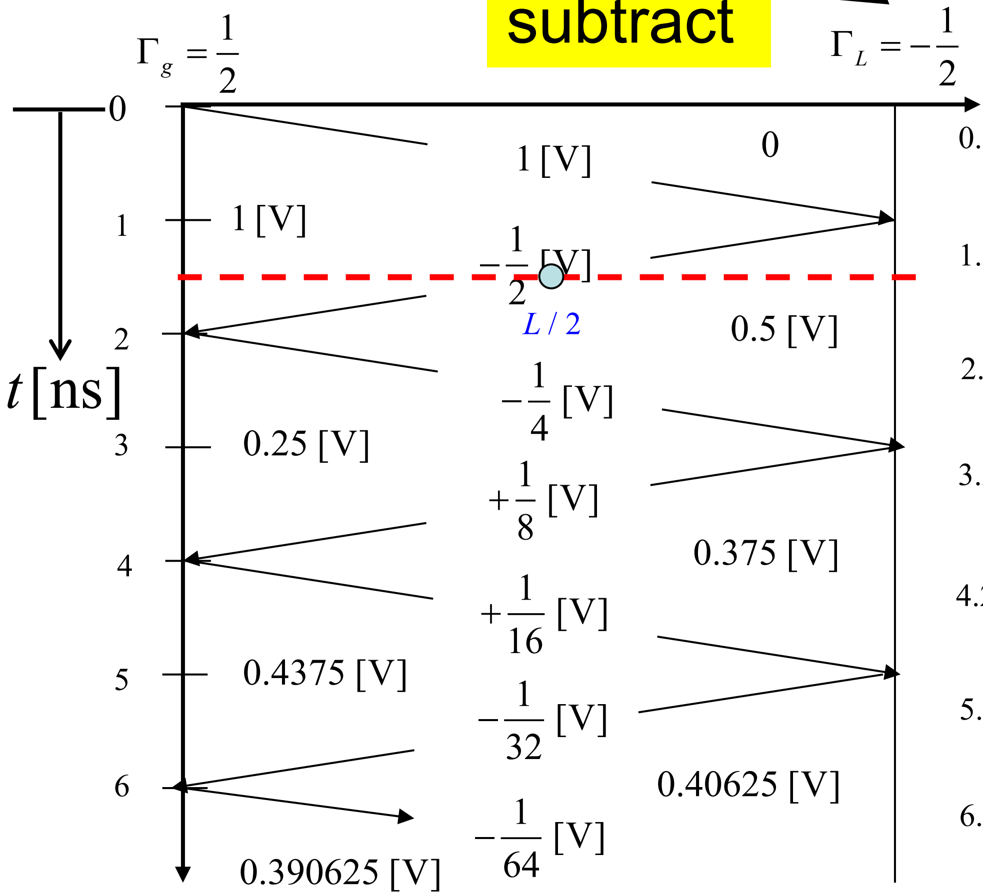



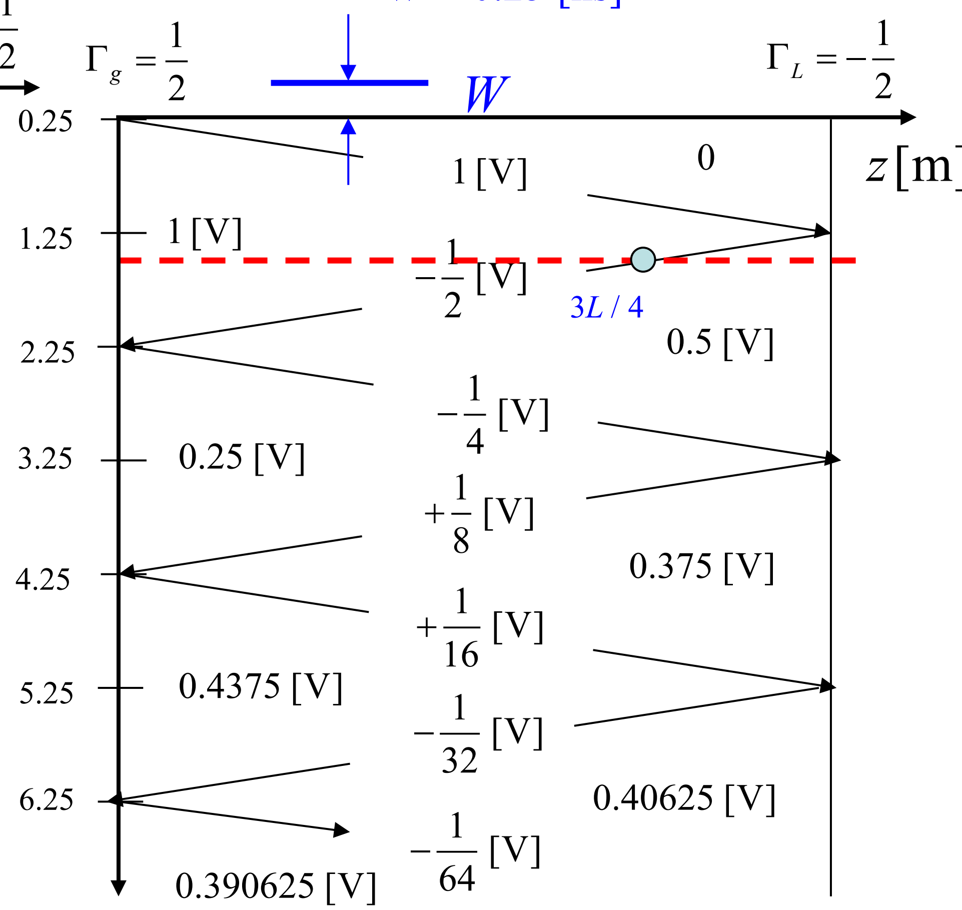



# Example: Pulse (cont.)

subtract

$W = 0.25$ [ns]




Snapshot of voltage

Snapshot

$t = 1.5$ [ns]